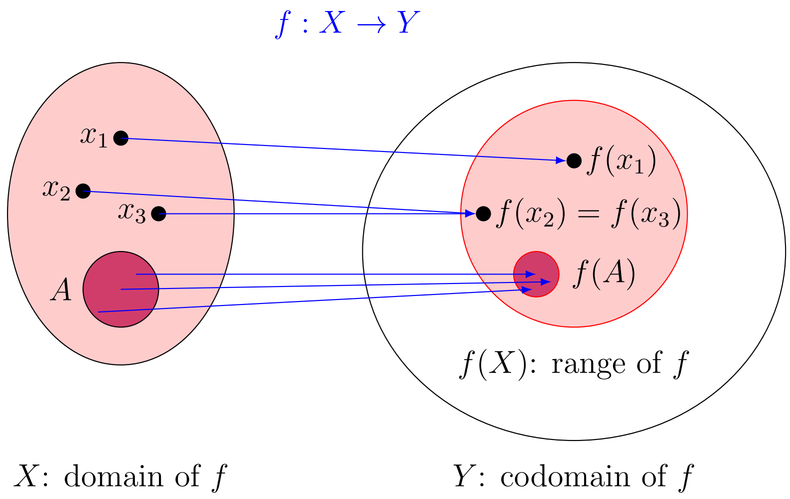
Topic 3: Functions[1]

The Language of Functions

We covered how, for any two sets X and Y, any ordered pair (x, y) associates an element $x \in X$ with an element $y \in Y$. Any collection of ordered pairs is considered a **binary relation** between X and Y.

A **function** is a special type of relation that associates each element of one set with a single, unique element of another set. Consider a relation f on $X \times Y$ whose elements are denoted as $(x, f(x))$. Such a relation is called a function if for each $x \in X$, there is *only one* $f(x)$ such that $(x, f(x)) \in f$.

You can imagine the function $f(x)$ is a machine that *whenever* you throw in the same x, it always gives $f(x)$ with no exception. Of course, it is possible that when you throw in some different xs, it gives you the same output. However, when you throw in any x, it cannot give out two different outputs.



We use $f : X \to Y$ to denote a function, which reads "f **is a mapping from** X **to** Y". The set X is called the **domain** of the function f, denoted as dom f or $\mathcal{D}(f)$. We also say "f **is defined on** X". The set Y is called the **codomain** of the function f.

[1]Instructors: Yue Yu and Sofia Olguin. This note was prepared for the 2026 UCSB Math Camp for Ph.D. students in economics. It incorporates materials from previous instructors, including Shu-Chen Tsao, ChienHsun Lin, and Sarah Robinson.

Each of the elements $x \in \text{dom } f$ corresponds to one and only one element y in the codomain. In which case, we say "f **maps** x **to** y", and the mapped element y is the **value** or the **image** of f at x, denoted as $f(x)$. The collection $\{(x, y) \in X \times Y | y = f(x)\}$ is called the **graph** of f.

We can also express the image of a subset of X. For any set $A \subset X$, the image of A under f, denoted $f(A)$, is the set of all images $f(x)$ for $x \in A$:

$$f(A) = \{f(x) \mid x \in A\}$$

Furthermore, the set of all values of f, $f(X)$, which is the image of the entire domain X, is called the **range of** f or the **image of** f. This is denoted as $\text{ran } f$ or $\mathcal{R}(f)$:

$$f(X) = \{f(x) \mid x \in X\}.$$

With this set/relation definition of functions, we can say that for any functions f and g, f and g are equal if the sets $f = g$.

Example 1

Consider functions $f : \mathbb{R}^2 \to \mathbb{R}$ and $g : \mathbb{R}^2 \to \mathbb{R}$ that

$$f(x, y) = |x - y| = \begin{cases} x - y & \text{if } x \geq y \\ y - x & \text{if } y > x \end{cases}$$

$$g(x, y) = \sqrt{(x - y)^2}$$

(a) What is the domain and the codomain of f and g?

(b) What is the range of f and g?

(c) Are f and g equal?

Based on how a function maps the elements, we can classify the function mappings.

Definition 1

A function $f : X \to Y$ is a **surjective function** or **onto function** if ran $f = Y$. That is,

$$\forall y \in Y,\ \exists x \in X \text{ such that } f(x) = y.$$

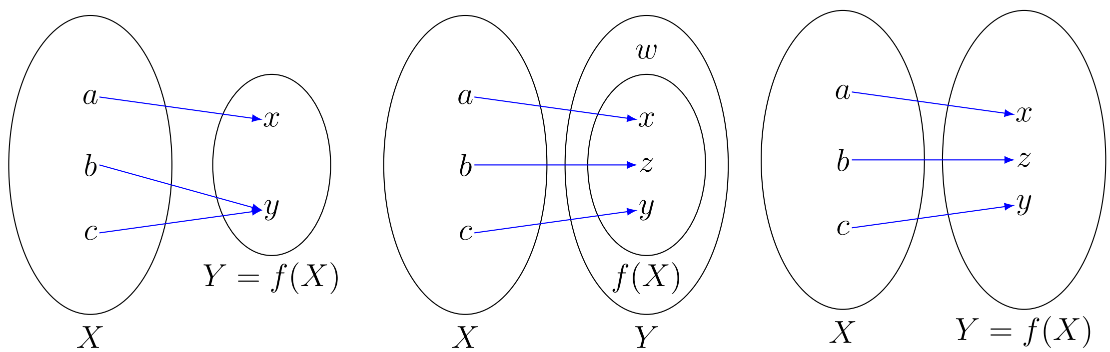
A function $f : X \to Y$ is a **injective function** or **one-to-one (1-1) function** if

$$\forall a, b \in X,\ \ f(a) = f(b) \Rightarrow a = b,$$

or equivalently,

$$\forall a, b \in X,\ \ a \neq b \Rightarrow f(a) \neq f(b).$$

A function $f : X \to Y$ is called a **bijective function** , **one-to-one correspondence**, or f is **1-1 onto** if f is surjective and injective.



surjective (not injective) — injective (not surjective) — bijective

Example 2

Check whether the following functions are surjective, injective, or bijective.

(1) $f : \mathbb{R} \to \mathbb{R},\ f(x) = x^2$

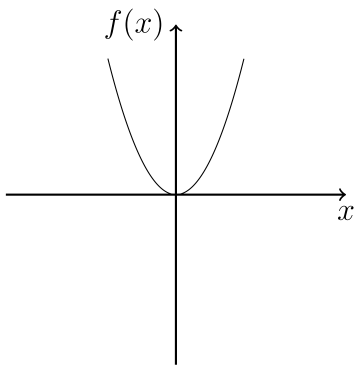
(2) $f : \mathbb{R}_{++} \to \mathbb{R},\ f(x) = \frac{1}{x}$

(3) $f : \mathbb{R} \to \mathbb{R},\ f(x) = (x+1)(x-1)(x-2) = x^3 - 2x^2 - x + 2$

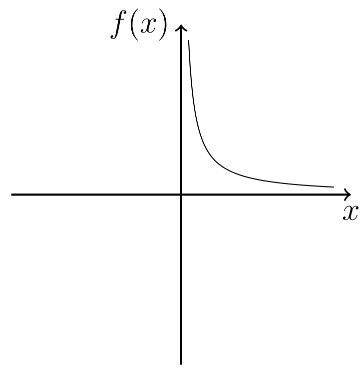
(4) $f : \mathbb{R}_{++} \to \mathbb{R},\ f(x) = \log x$

Note: Sometimes we use $\mathbb{R}_{++}$ to denote $\mathbb{R}_+$ that explicitly excludes zero.

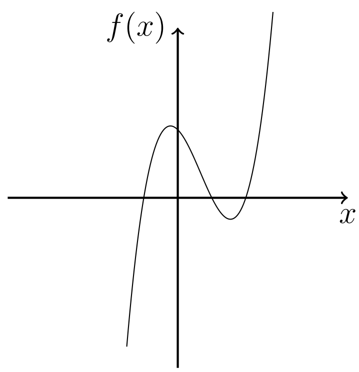
We can check if they are surjective, injective, or bijective by looking at the graph. But you should also practice how to formally prove them using what we learned in previous lectures.



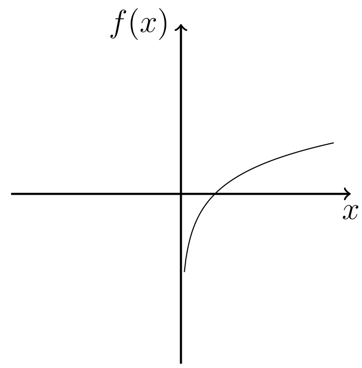
$f(x) = x^2$



$f(x) = \frac{1}{x}$



$f(x) = x^3 - 2x^2 - x + 2$



$f(x) = \log x$

Function Operation

Here are some notations for function operations.

Definition 2

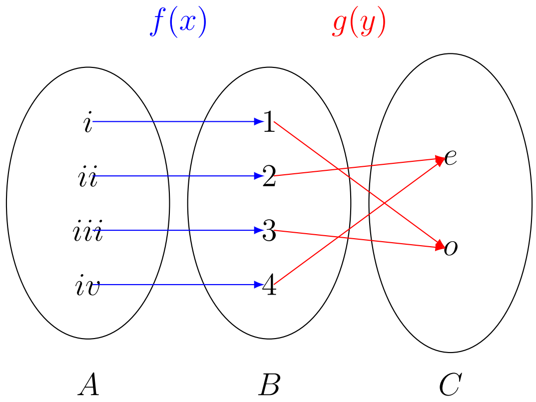
Let $f : X \to Y$ and $g : X \to Y$ be functions.

(1) $(f + g)(x)$= $f(x) + g(x)$ is the function of the sum of the functions f and g at each x.

(2) $(fg)(x)$= $f(x) \cdot g(x)$ is the function of the product of the functions f and g at each x.

Function Composition

Under some conditions, we can also *transmit* the functions. For example, let $A = \{i, ii, iii, iv\}$, $B = \{1, 2, 3, 4\}$, and $C = \{e, o\}$. Also let $f : A \to B$, $g : B \to C$.



Since the range of f is a subset of the domain of g, for every element $x \in A$, we can put in the function value $f(x)$ into g. Formally,

$$(g \circ f)(x) = g(f(x)).$$

The function $g \circ f$ is called the **composition** of f and g, read as "g **of** f" or "g composed with f". The domain of $g \circ f$ is A, and its codomain is C.

In general, $f \circ g \neq g \circ f$. In fact, in the example above, $f \circ g$ cannot exist, as the range of g is not a subset of f.

Proposition 1

Let $f : A \to B$ and $g : B \to C$.

(1) If f and g are surjective, then so is $g \circ f$.
(2) If f and g are injective, then so is $g \circ f$.
(3) If f and g are bijective, then so is $g \circ f$.

Proof. We show (1) first. Pick any $z \in C$. Since g is surjective, there is $y \in B$ such that $g(y) = z$. Furthermore, since f is also surjective, for that particular y, there exists $x \in A$ such that $f(x) = y$. Combining the equations we can get $g(f(x)) = z$. Since this is true for any $z \in C$, $g \circ f$ is surjective.
Then we show (2). Pick $x_0, x_1 \in A$, and $x_0 \neq x_1$. Since f is injective, $f(x_0) \neq f(x_1)$. Also, because g is injective, $g(f(x_0)) \neq g(f(x_1))$. By definition, $(g \circ f)(x_0) \neq (g \circ f)(x_1)$. Since this is true for any $x_0, x_1 \in A$, $g \circ f$ is injective.
Since $g \circ f$ is surjective and injective, $g \circ f$ is bijective. ■

We can also have multiple layers of compositions, and function composition is associative.

Proposition 2

For nonempty sets X, Y, Z, and W, let $f : X \to Y$, $g : Y \to Z$, and $h : Z \to W$. Then

$$(h \circ g) \circ f = h \circ (g \circ f).$$

Proof. First note that both $(h \circ g) \circ f$ and $h \circ (g \circ f)$ are mappings from X to W. Pick $x \in X$. Let $f(x) = y$, $g(y) = z$ and $h(z) = w$. Then

$$(h \circ g) \circ f = (h \circ g)(f(x)) = (h \circ g)(y) = h(g(y)) = h(z) = w,$$

and

$$h \circ (g \circ f)(x) = h((g \circ f)(x)) = h(g(f(x))) = h(g(y)) = h(z) = w.$$

Since $x \in X$ is arbitrarily chosen, the function values of $(h \circ g) \circ f$ and $h \circ (g \circ f)$ always the same. Therefore $(h \circ g) \circ f = h \circ (g \circ f)$. ■

Inverse Function

For a relation R from X to Y, the **inverse relation** of R is defined as

$$R^{-1} = \{(y, x) \mid (x, y) \in R\}.$$

For example, consider the relation $R = \{$(apple, red), (banana, yellow), (lemon, yellow)$\}$, the inverse relation of R is $R^{-1} = \{$(red, apple), (yellow, banana), (yellow, lemon)$\}$. Since a function is also a relation, we can find the inverse relation of the function. However, the inverse relation of the function is not necessarily a function. For example. R is a function, but R^{-1} is not a function. Loosely speaking, if we throw "yellow" into R^{-1}, it gives us both "banana" and "lemon," which violates the definition of a function.

Now we introduce the condition that ensures that the inverse relation of a function is also a function.

Proposition 3

Let $f : X \to Y$ be a function. The inverse relation of the function f, denoted as $f^{-1} : Y \to X$, is also a function if and only if f is bijective. We call f^{-1} the **inverse function** of f.

The proof is quite long, so we omitted it here. Since the proposition shows an equivalence, the general strategy is to show that f^{-1} is a function implies f is bijective, and then show that if f is bijective then f^{-1} is a function.

Corollary

Suppose $f : X \to Y$ has an inverse function $f^{-1} : Y \to X$. Then

(1) f^{-1} is also bijective,

(2) the inverse of f^{-1} exists and $(f^{-1})^{-1} = f$, and

(3) $f^{-1} \circ f(x) = f^{-1}(f(x)) = x$ for any $x \in X$.

The last statement is a useful property of inverse functions: the composition of a function and its inverse function maps an element back to itself. We can then use this property to verify whether two functions are inverse to each other.

Example 3

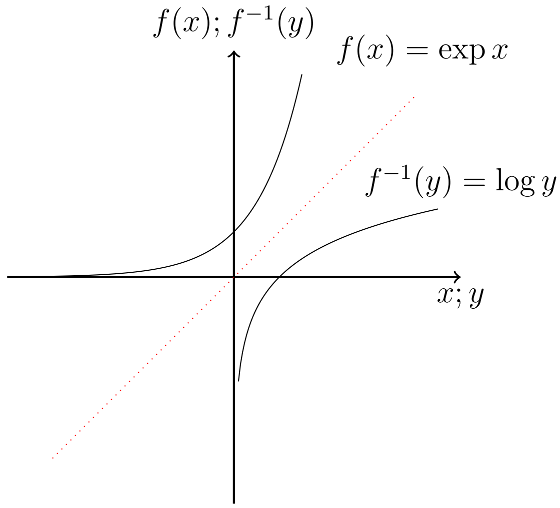
Find the inverse function of the following functions.

(1) $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x$

(2) $f : \mathbb{R} \to \mathbb{R}_{++}$, $f(x) = \exp(x)$

(3) $f : \mathbb{R} \setminus \{0\} \to \mathbb{R} \setminus \{0\}$, $f(x) = \frac{1}{x}$

For $f : \mathbb{R} \to \mathbb{R}$, if we draw the function on the plain, we will see that f and f^{-1} will be symmetric with respect to the 45 degrees line, $y = x$.



Monotonic functions

Real functions are defined as those with domains and ranges in $\mathbb{R}$. This is the main character for real analysis and the most frequently seen family of functions in economics. We will go over some of the properties of real functions.

Definition 3 (Increasing and Decreasing Functions)

We call $f : \mathbb{R} \to \mathbb{R}$ an **increasing (decreasing) function** if

$$\text{whenever } x_1 > x_0,\ f(x_1) \geq (\leq) f(x_0).$$

In this case, we say $f(x)$ is increasing (decreasing) in x.

We call $f : \mathbb{R} \to \mathbb{R}$ an **strictly increasing (decreasing) function** if

$$\text{whenever } x_1 > x_0,\ f(x_1) > (<) f(x_0).$$

In this case, we say $f(x)$ is strictly increasing (decreasing) in x.

Definition 4 (Monotonic functions)

We say the function $f(x)$ is **monotonic** if $f(x)$ is either increasing or decreasing.

We say the function $f(x)$ is **strictly monotonic** if $f(x)$ is either strictly increasing or strictly decreasing.

Sometimes functions may not be always increasing or decreasing. We say a function is **increasing in the (open or closed) interval** I if for any $x_1, x_0 \in I$, whenever $x_1 > x_0$, $f(x_1) \geq f(x_0)$.

Monotonic functions in $\mathbb{R}^n$

We can also define functions on $\mathbb{R}^n$. A function $f : \mathbb{R}^n \to \mathbb{R}$ is called a **multivariate real function**. We use the boldface notation, such as $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$, to denote vectors in n-dimensional real space. To compare such vectors, we introduce an ordering on $\mathbb{R}^n$.

Definition 5 (Vector order)

Let $\mathbf{x},\ \mathbf{y} \in \mathbb{R}^n$. Then we denote

- $\mathbf{x} \geq \mathbf{y}$ if $x_i \geq y_i$ for every $i = 1, \dots, n$.
- $\mathbf{x} \gg \mathbf{y}$ if $x_i > y_i$ for every $i = 1, \dots, n$.

Then we can have a version of monotonic functions for the multivariate functions.

Definition 6 (Monotonic functions in $\mathbb{R}^n$)

We call $f : \mathbb{R}^n \to \mathbb{R}$ an **increasing function** if

$$\text{whenever } \mathbf{x} \geq \mathbf{y},\ f(\mathbf{x}) \geq f(\mathbf{y}).$$

We call $f : \mathbb{R}^n \to \mathbb{R}$ an **strictly increasing function** if

$$\text{whenever } \mathbf{x} \gg \mathbf{y},\ f(\mathbf{x}) > f(\mathbf{y}).$$

We call $f : \mathbb{R}^n \to \mathbb{R}$ an **strongly increasing function** if

$$\text{whenever } \mathbf{x} \neq \mathbf{y} \text{ and } \mathbf{x} \geq \mathbf{y},\ f(\mathbf{x}) > f(\mathbf{y}).$$

Example 4

Consider $u : \mathbb{R}^2 \to \mathbb{R} \quad u(x_1, x_2) = \min\{x_1, x_2\}$.
Is u increasing? Is u strictly increasing? Is u strongly increasing?

Example 5

If f is strictly increasing, is it strongly increasing? If f is strongly increasing, is it strictly increasing? Prove your results.

Homogeneous and Homothetic Functions

Homogeneous and homothetic functions are frequently used in economics.

A function f is called **homogeneous of degree** r if

$$f(tx_1, tx_2, ..., tx_n) = t^r f(x_1, x_2, ..., x_n)$$

for every $x_1, x_2, ..., x_n$ and $t > 0$.

Homogeneous functions exhibit very regular behavior as all variables are increased simultaneously and in the same factor (t). Two special cases are particularly noteworthy:

- A function is **homogeneous of degree one** or linearly homogeneous if $f(t\mathbf{x}) = tf(\mathbf{x})\ \forall t > 0$. In this case, doubling or tripling all variables doubles or triples the function's value.
- A function is **homogeneous of degree zero** if $f(t\mathbf{x}) = f(\mathbf{x})\ \forall t > 0$. Here, proportional changes in all variables leave the function's value unchanged.

Example 6

For each of the following functions, determine whether it is homogeneous. If so, specify the degree of homogeneity.

(a) $f(x_1, x_2) = x_1^2 + x_2^2$.

(b) Cobb-Douglas production function: $f(K, L) = AK^\alpha L^\beta$. What if $\beta = 1 - \alpha$?

A function f is called **homothetic** if there exists a continuous and strictly monotonic function g and a homogeneous function h such that $f = g \circ h$.

Note that a homogeneous function is always a homothetic function. We can see this simply by letting $g(a) = a$. However, a homothetic function is not necessarily homogeneous.

Correspondence

Sometimes, we still need to consider relations more than functions. For example, consider the set

$$M = \{\text{McDonald's, KFC, Wendy's, Chick-Fil-A}\}.$$

Jane wants to pick her favorite fast food restaurants from the list. However, she would rank both Wendy's and Chick-Fil-A on the top.

If we just use a function $P : A \to M$ to describe the favorite fast food restaurant and A is the set of people, there will be a violation to the definition of the functions as $P(\text{Jane})$ is not unique. Therefore, we define a correspondence that can give us a set of values. A correspondence is common in the theory of preference in Econ 210A.

Definition 7 (Correspondence)

Let X and Y be sets. A relation ϕ is a **correspondence**, denoted as $\phi : X \rightrightarrows Y$, if for every $x \in X$, $\phi(x) \subset Y$.

ϕ is **well-defined** if for every $x \in X$, $\phi(x)$ is *non-empty*.

The **image** of ϕ is defined and denoted as follows:

$$\text{Im}_\phi(x) = \{y \in Y | y \in \phi(x)\}.$$

If for every x, $\text{Im}_\phi(x) \subset \mathbb{R}^n$, then ϕ is a **real-valued** correspondence.

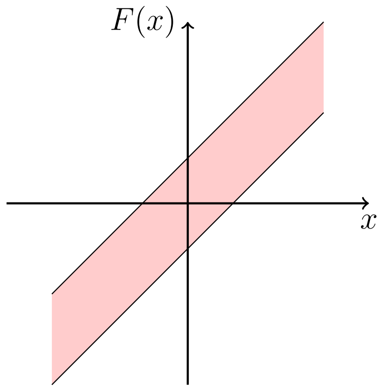
In the example of fast food choice, let $P : A \rightrightarrows M$. Then

$$P(\text{Jane}) = \{\text{Wendy's, Chick-Fil-A}\}.$$

Example 7

Draw the graphs of the following correspondences on $\mathbb{R}$.

- $F(x) = [x-a, x+a]$ for some $a \in \mathbb{R}$
- $F(x) = \begin{cases} [0.4, 0.6] & \text{if } x \leq 1 \\ 0.5 & \text{o.w.} \end{cases}$
- $F(x) = \begin{cases} [0.4, 0.6] & \text{if } x < 1 \\ 0.5 & \text{o.w.} \end{cases}$



$F(x) = [x-a, x+a]$ for some $a \in \mathbb{R}$

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